Contents

Canonical Morphism Sheaves $G \to f_* f^{-1} G$ **1**

Canonical Morphism Sheaves $G \to f_* f^{-1} G$

We construct

$$\eta_G : G \to f_* f^{-1} G$$

For each $s \in G(V)$, define compatible family of sections over $f^{-1}(V)$.

$$\forall U \subseteq f^{-1}(V) \qquad s_U := s|_{f(U)} \in G(f(U))$$

These assemble into a section of $f^{-1}G$ over $f^{-1}(V)$, thus we get a natural map

$$\eta_{G,V} : G(V) \to f_* f^{-1} G(V)$$

which commutes on restrictions.